Conducting Forensic Investigations on Linux Systems (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 06

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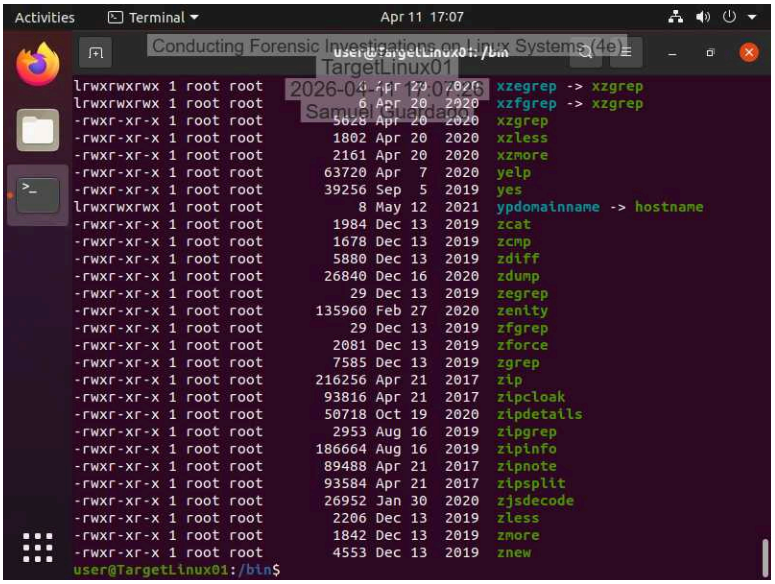
Report Generated: Thursday, July 30, 2026 at 6:30 PM

Time on Task: 1 hour, 26 minutes
Progress: 100%

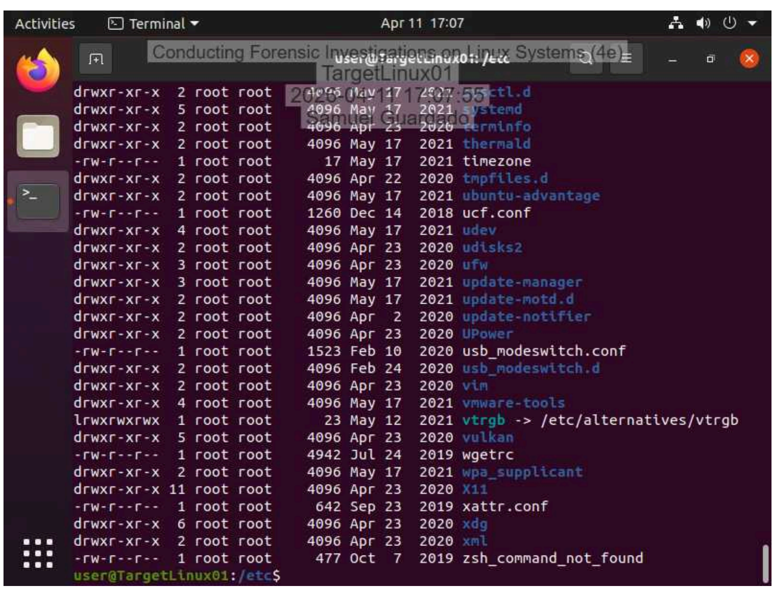
Section 1: Hands-On Demonstration

Part 1: Explore a Live Linux System

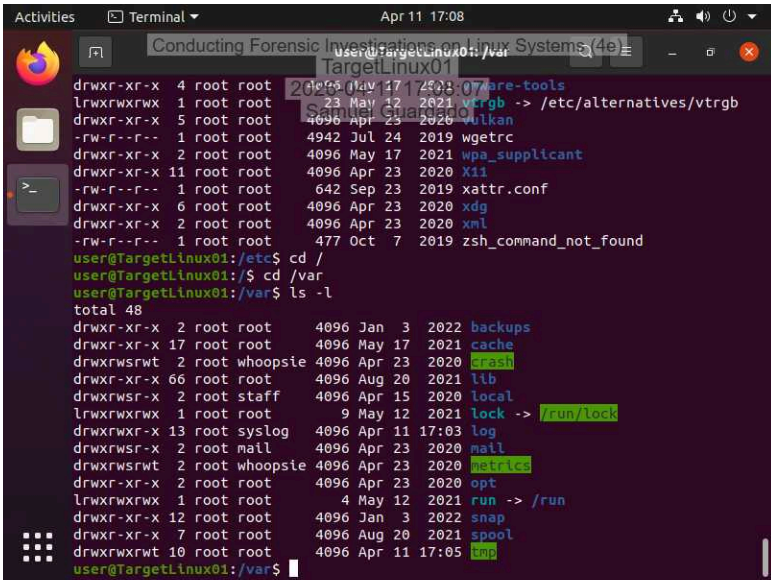
17. **Make a screen capture** showing the **contents of the /bin directory**.



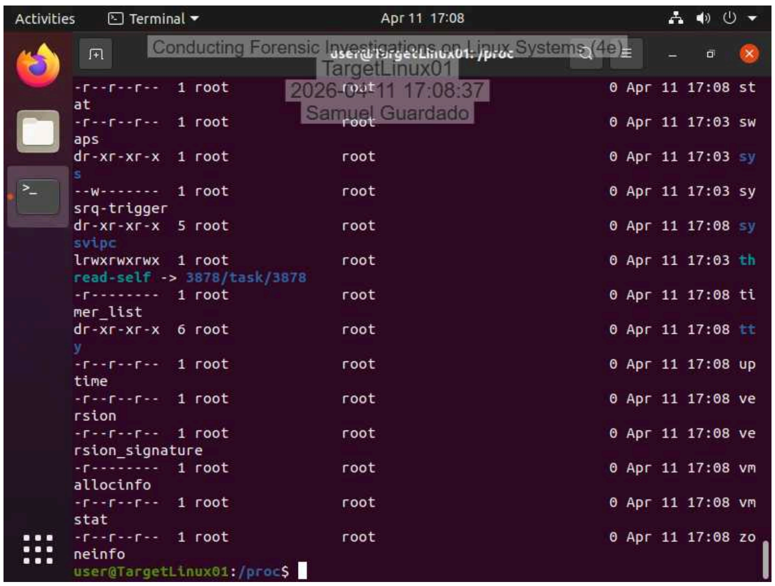
20. **Make a screen capture** showing the **contents of the /etc directory**.



21. **Make a screen capture** showing the **contents of the /var directory**.

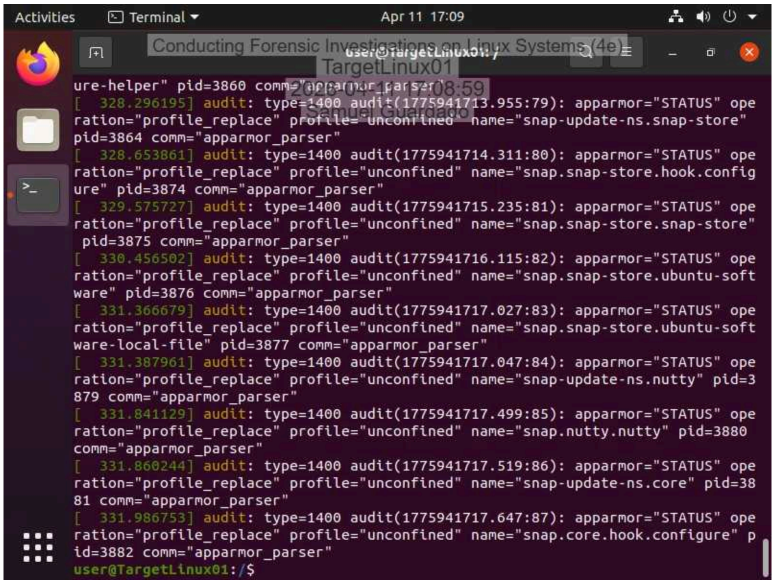


22. **Make a screen capture** showing the **contents of the /proc directory**.

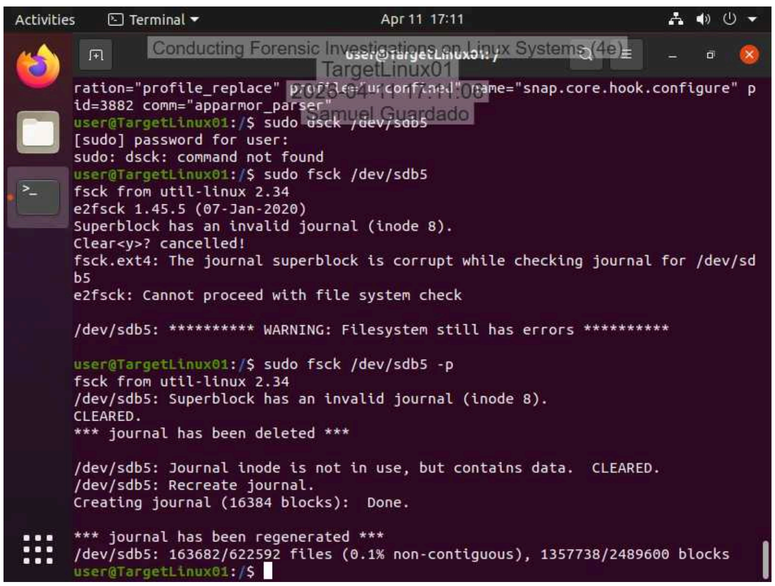


Part 2: Use Linux Shell Commands for Forensic Investigations

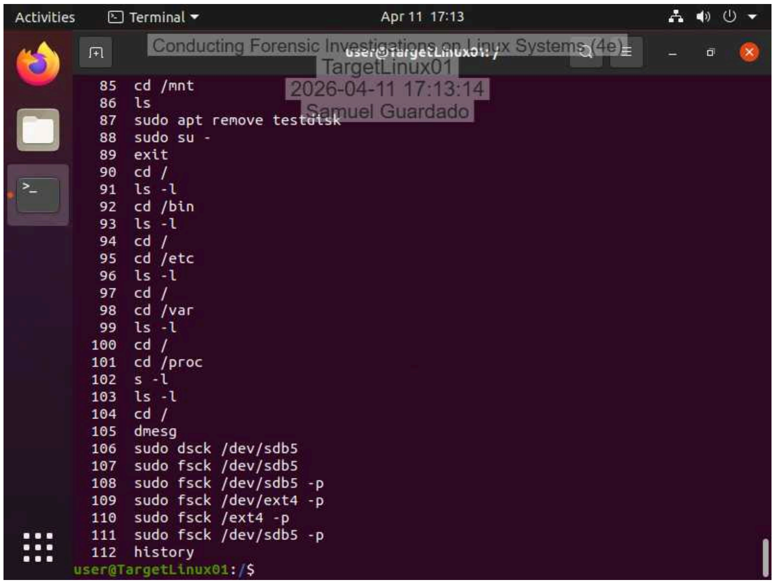
2. **Make a screen capture** showing the **results of the dmesg command**.



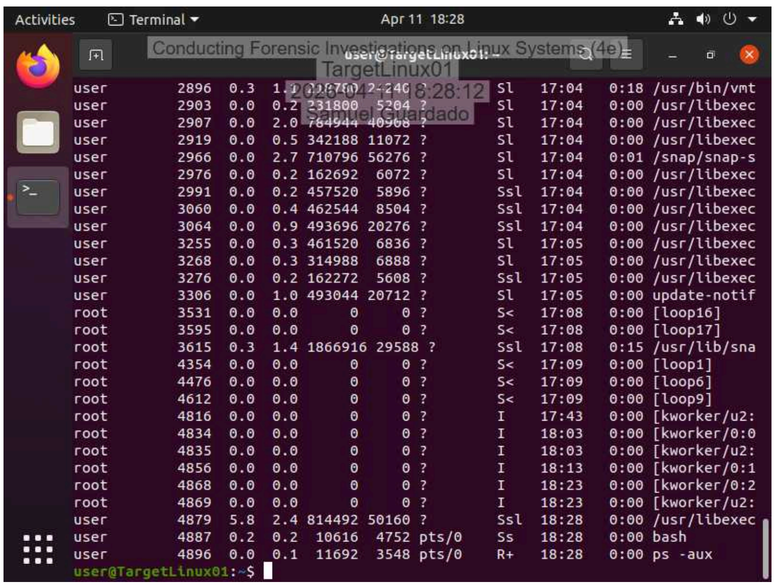
7. **Make a screen capture** showing the **results of the fsck command.**



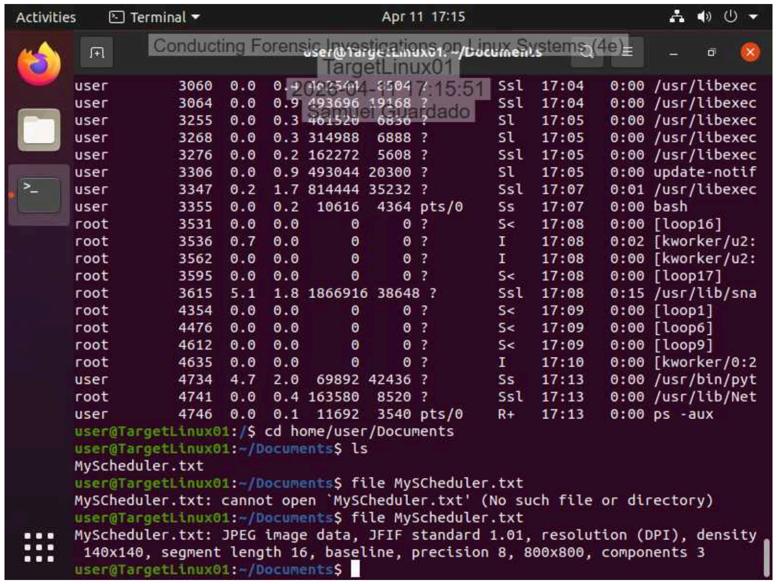
9. **Make a screen capture** showing the **results of the history command**.



11. **Make a screen capture** showing the **running processes**.

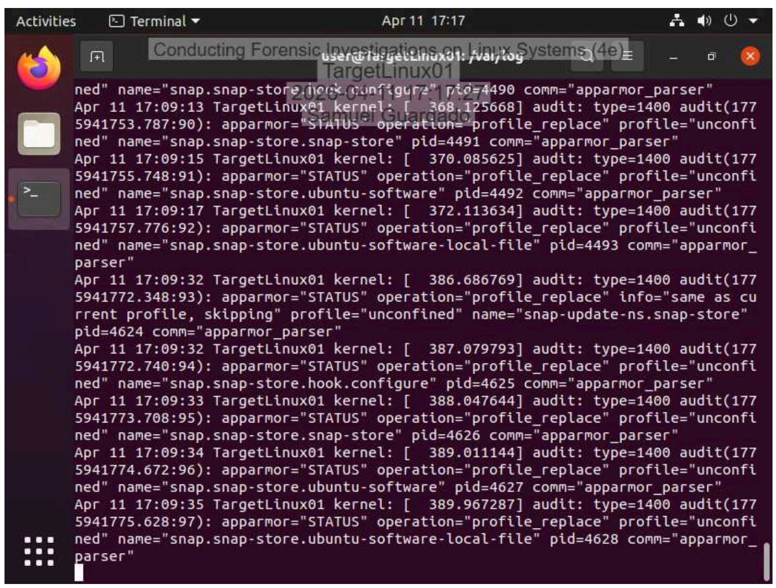


15. **Make a screen capture** showing the **results of the file command**.

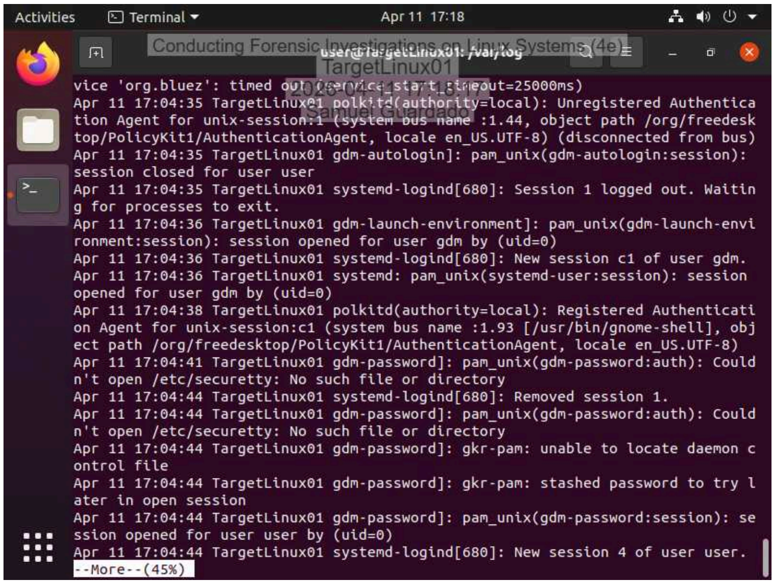


Part 3: Retrieve Logs Files on a Live Linux System

4. **Make a screen capture** showing the **records in the kern.log file.**



7. **Make a screen capture** showing the **records in the auth.log file**.



Section 2: Applied Learning

Part 1: Identify Login Attempts on a Linux Drive Image

15. **Document** the names of the two non-root users that attempted to log in, the number of attempts detected, the date/time range of the attempts, the source IP address for the login attempts, and the port.

The two non root users were Dominic, and Noel. With 22 attempts, with the source IP: 192.78.1 on port 2511 on the 11th of June, with a time range of 00:57:19-05:39:01

17. **Document** the date and time the most recent successful login for the user(s) that you previously identified in step 15.

Noel was not able to get into the system, on the other hand, Dominic's last successful login was on June 11th, at 05:23:03.

Part 2: Identify Software Installations on a Linux Drive Image

3. **Document** the applications that were installed using apt-get, then use the Internet to identify the ones that might be considered suspicious.

Logkeys, autotools-dev, autoconf, kbd, cacti, and openssh.serverSuSpcicious activity: Logkeys. Openssh casn be used for remote access, and possibly gain back door access. Cacti is a network monitoring tool used by attackers to gain unauthorized access to networks, execute remote commands, and steal data.

Part 3: Identify External Drive Attachments on a Linux Drive Image

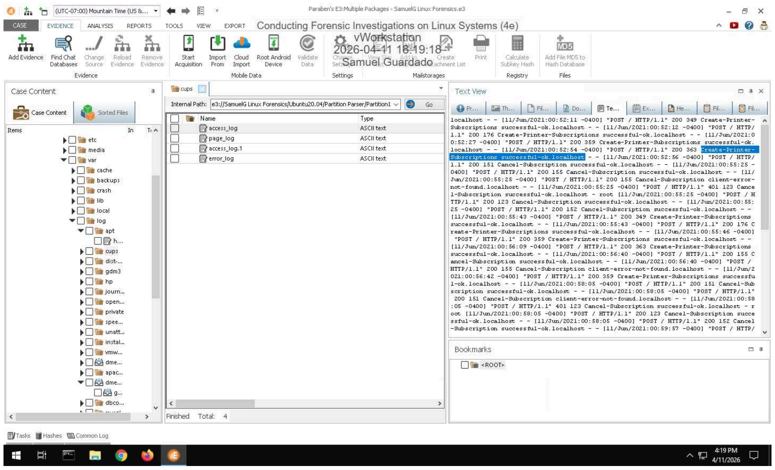
4. **Document** when the USB storage device was connected and its serial number.

It was detected on Jun 10 at 10:24:12. Serial Number: FBI1405291710344

Section 3: Challenge and Analysis

Part 1: Identify Recently Printed Files on a Linux Drive Image

Make a screen capture showing the **contents of the printer log file**.



Part 2: Identify Disk Imaging on a Linux Drive Image

Make a screen capture showing the **record of the dd command in the Text View**.